We can identify two functions, one for calculating the area of a circle and the other for the circumference of the circle. If we abstract the two functions as circle_area(r) and circle_circumference(r), then cylinder_area(r, h) can be solved as

cylinder_area (r,h) = 2 X circle_area (r) + circle_circumference (r) X h

6.4 Algorithm Design Techniques

There are a few basic principles and techniques for designing algorithms.

1. **Specification:** The first step in problem solving is to state the problem precisely. A problem is specified in terms of the input given and the output desired. The specification must also state the properties of the given input, and the relation between the input and the output.
2. **Abstraction:** A problem can involve a lot of details. Several of these details are unnecessary for solving the problem. Only a few details are essential. Ignoring or hiding unnecessary details and modeling an entity only by its essential properties is known as abstraction. For example, when we represent the state of a process, we select only the variables essential to the problem and ignore inessential details.
3. **Composition:** An algorithm is composed of assignment and control flow statements. A control flow statement tests a condition of the state and, depending on the value of the condition, decides the next statement to be executed.
4. **Decomposition:** We divide the main algorithm into functions. We construct each function independently of the main algorithm and other functions. Finally, we construct the main algorithm using the functions. When we use the functions, it is enough to know the specification of the function. It is not necessary to know how the function is implemented.

6.5 Specification

To solve a problem, first we must state the problem clearly and precisely. A problem is specified by the given input and the desired output. To design an algorithm for solving a problem, we should know the properties of the given input and the properties of the desired output. The goal of the algorithm is to establish the relation between the input and the desired output.

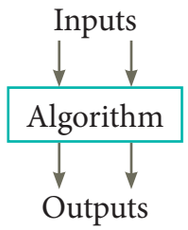


Figure 6.1: Input-output relation

An algorithm is specified by the properties of the given input and the relation between the input and the desired output. In simple words, specification of an algorithm is the desired input-output relation.

The inputs and outputs are passed between an algorithm and the user through variables. The values of the variables when the algorithm starts is known as the initial state, and the values of the variables when the algorithm finishes is known as the final state.

Let P be the required property of the inputs and Q the property of the desired outputs. Then the algorithm S is specified as

Chapter 6 Page 076-087.indd 79 3/24/2020 9:14:09 AM